Akash Maiti

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Education

The University of Texas at Austin May 2028
B.S. Electrical and Computer Engineering (Honors) *Austin, TX*
- Honors: University Honors, T.W. Whaley Friends of Alec Endowed Scholarship ($36,000 Total Award Value)
- Coursework: Information Security, Software Testing, Software Design, Data Structures, Algorithms

Technical Skills

Web & App Security: OWASP Top 10, XSS, SQLi, SSRF, IDOR, CSRF, Secure Code Review, Burp Suite
AI for Security: LLM Agents, Agentic Workflows, Synthetic Data, PyTorch, TensorFlow, Anomaly Detection
Languages: Python, JavaScript/Node.js, Java, C/C++, SQL, Bash, PowerShell, PHP
Infrastructure: AWS (Lambda, IAM), Linux, Windows, Active Directory, Splunk (SIEM), Wireshark, TCP/IP
Certifications: CompTIA Security+ (Exp. Jan 2027), HTB CDSA, Splunk Core Certified User (Exp. Nov 2026)

AI Security Projects & Competitions

vulnscan: LLM-Assisted Vulnerability Scanner | *Python, Semgrep, Pysa, Local LLM* 2026
- Built an AI-driven static analysis pipeline feeding Semgrep and Pysa taint findings to a local LLM for automated triage.
- Engineered LLM triage that filtered false-positive vulnerability findings on real apps, lifting precision from **38%** to **50%**.
- Tuned the triage to suppress noise without losing real bugs, retaining **91%** of true-positive vulnerabilities.
- Scored results against ground truth from vulnerable apps, a synthetic benchmark, and real CVEs across **XX** findings.
- Ran inference on a local model to keep scanned source code private, publishing methodology and results as open research.

Collegiate Cyber Defense Competition (CCDC) | *Business & IR Lead, HASH* Jan 2025 – Present
- Defended Windows/Linux systems against live red teams, winning Southwest Regionals and placing **5th nationally**.
- Automated host and network hardening with PowerShell and Bash, securing **8** devices and **15+** microservices.
- Directed incident response and business injects under active attack, delivering executive breach reports over **16** hours.

Navigo | *Software Lead – LLMs, Agentic Code Automation, Sensitive Health Data* Sept 2024 – Present
- Deployed autonomous agentic coding systems that detect and self-heal bugs, reducing transcription error rates by **30%+**.
- Architected an AI MVP applying LLM summarization to sensitive patient data, cutting clinical charting time by **50%**.
- Interviewed **50+** regional healthcare stakeholders to validate the pilot, securing **$1,250** in grant funding.

Experience

University of Texas at Austin, *Regional SOC Analyst* March 2025 – Present
- Built agentic AI tooling for automated threat detection and alert triage, cutting weekly manual analyst work by **60%+**.
- Traced adversary attack chains in audit and network logs to MITRE ATT&CK, remediating **20+** vulnerabilities.
- Investigated and escalated security incidents across university critical infrastructure using Splunk SIEM log correlation.
- Mastered incident response, network forensics, and threat hunting through a selective **700-hour** cyber defense program.

DevRev, *Security Engineering Intern* Jan 2026 – Aug 2026
- Engineered agentic workflows and synthetic data to harden production LLM chatbots, cutting hallucinations by **50%+**.
- Isolated untrusted code in AWS Lambda sandboxes with least-privilege IAM, reducing lateral movement risk by **40%**.
- Automated global security training infrastructure in Python, sustaining weekly sessions for **200+** employees.

Strauss Center for International Security and Law, *Cybersecurity Risk Consultant* Aug 2025 – May 2026
- Led vulnerability assessments of a mental health provider's patient data systems, uncovering **30+** security risks.
- Authored risk-prioritized remediation plans and security policies, achieving **100%** HIPAA-aligned clinical workflows.

Trace Labs, *CTF Challenge Creator* Sept 2026 – Present
- Designed OSINT capture-the-flag challenges for a global security community event, engaging **7,000+** participants.
- Redesigned challenge interactivity and difficulty progression, boosting global participant engagement by **20%**.

Leadership

Information and Systems Security Society (ISSS), *Corporate Co-Director* Jan 2025 – Present
- Directed outreach to CrowdStrike, Sandia, Lockheed Martin, and other security firms, securing **$5,000+** in sponsorships.

Longhorn Developers, *Technical Lead* June 2025 – Present
- Led development of an A*-routed campus navigation app for students with disabilities, managing **7** developers.
- Automated accessibility data integration with UT Mapping, reducing average student navigation times by **20%+**.